

Thực hành KTMT Lab 13

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Assignment 4

- Code

```
.eqv IN_ADRESS_HEXА_KEYBOARD 0xFFFF0012
.eqv OUT_ADRESS_HEXА_KEYBOARD 0xFFFF0014
.eqv COUNTER 0xFFFF0013 # Time Counter
.eqv MASK_CAUSE_COUNTER 0x00000400 # Bit 10: Counter interrupt
.eqv MASK_CAUSE_KEYMATRIX 0x00000800 # Bit 11: Key matrixinterrupt
.data
msg_keypress: .asciiz "Someone has pressed a key: "
msg: .asciiz "\n"
msg_counter: .asciiz "Time inteval!\n"
#~~~~~
# MAIN Procedure
#~~~~~
.text
main:
#-----
# Enable interrupts you expect
#-----
# Enable the interrupt of Keyboard matrix 4x4 of Digital Lab Sim
li $t1, IN_ADRESS_HEXА_KEYBOARD
li $t3, 0x80 # bit 7 = 1 to enable
sb $t3, 0($t1)
# Enable the interrupt of TimeCounter of Digital Lab Sim
```

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li $t1, COUNTER
sb $t1, 0($t1)

#-----

# Loop an print sequence numbers

#-----

Loop: nop
nop
nop
sleep: addi $v0,$zero,32 # BUG: must sleep to wait for Time Counter
li $a0,200 # sleep 300 ms
syscall
nop # WARNING: nop is mandatory here.
b Loop
end_main:

#~~~~~

# GENERAL INTERRUPT SERVED ROUTINE for all interrupts

#~~~~~

.ktext 0x80000180
IntSR: #-----

# Temporary disable interrupt

#-----

dis_int:li $t1, COUNTER # BUG: must disable with Time Counter
sb $zero, 0($t1)

# no need to disable keyboard matrix interrupt

#-----

# Processing

#-----

get_caus:mfc0 $t1, $13 # $t1 = Coproc0.cause

```

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IsCount:li $t2, MASK_CAUSE_COUNTER# if Cause value confirm Counter..
and $at, $t1,$t2
beq $at,$t2, Counter_Intr
IsKeyMa:li $t2, MASK_CAUSE_KEYMATRIX # if Cause value confirm Key..
and $at, $t1,$t2
beq $at,$t2, Keypress_Intr
others: j end_process # other cases
Keypress_Intr: li $v0, 4 # Processing Key Matrix Interrupt
la $a0, msg_keypress
syscall
#-----
# Kiểm tra xem nút nào được bấm
li $t1, IN_ADDRESS_HEX_KEYBOARD
li $t2, OUT_ADDRESS_HEX_KEYBOARD
li $t3, 0x01 # check row 4 with key 0, 1, 2, 3
li $t4, 0x02 # check row 4 with key 4, 5, 6, 7
li $t5, 0x04 # check row 4 with key 8, 9, A, B
li $t6, 0x08 # check row 4 with key C, D, E, F
sb $t3, 0($t1) # must reassign expected row
lbu $a0, 0($t2) # read scan code of key button
bne $a0, $zero, print
sb $t4, 0($t1) # must reassign expected row
lbu $a0, 0($t2) # read scan code of key button
bne $a0, $zero, print
sb $t5, 0($t1) # must reassign expected row
lbu $a0, 0($t2) # read scan code of key button
bne $a0, $zero, print
sb $t6, 0($t1) # must reassign expected row

```

```

lbu $a0, 0($t2) # read scan code of key button
bne $a0, $zero, print
#-----

print:
li $v0, 34 # print integer (hexa)
syscall
print_n:
li $v0, 4 # Processing Counter Interrupt
la $a0, msg
syscall
#-----
# Re-enable the interrupt of Keyboard matrix 4x4 of Digital Lab Sim
li $t2, IN_ADRESS_HEX_A_KEYBOARD
li $t3, 0x80 # bit 7 = 1 to enable
sb $t3, 0($t2)
#-----
j end_process
Counter_Intr: li $v0, 4 # Processing Counter Interrupt
la $a0, msg_counter
syscall
j end_process
end_process:
mtc0 $zero, $13 # Must clear cause reg
en_int: #-----
# Re-enable interrupt
#-----
li $t1, COUNTER
sb $t1, 0($t1)

```

```

#-----
# Evaluate the return address of main routine
# epc <= epc + 4
#-----
next_pc:mfc0 $at, $14 # $at <= Coproc0.$14 = Coproc0.epc
addi $at, $at, 4 # $at = $at + 4 (next instruction)
mtc0 $at, $14 # Coproc0.$14 = Coproc0.epc <= $at
return: eret # Return from exception

```

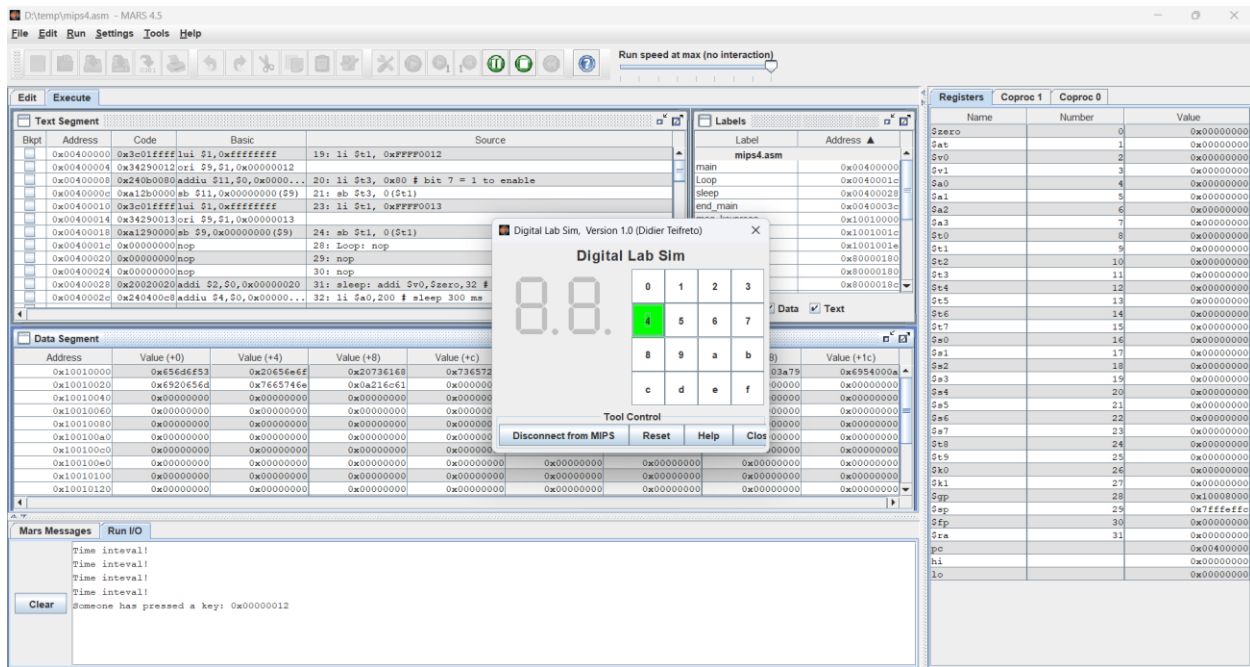
- Thực thi

-Khi nhảy đoạn mã cho xử lý ngắt, kiểm tra xem đó là ngắt của Counter hay Keyboard matrix 4x4.

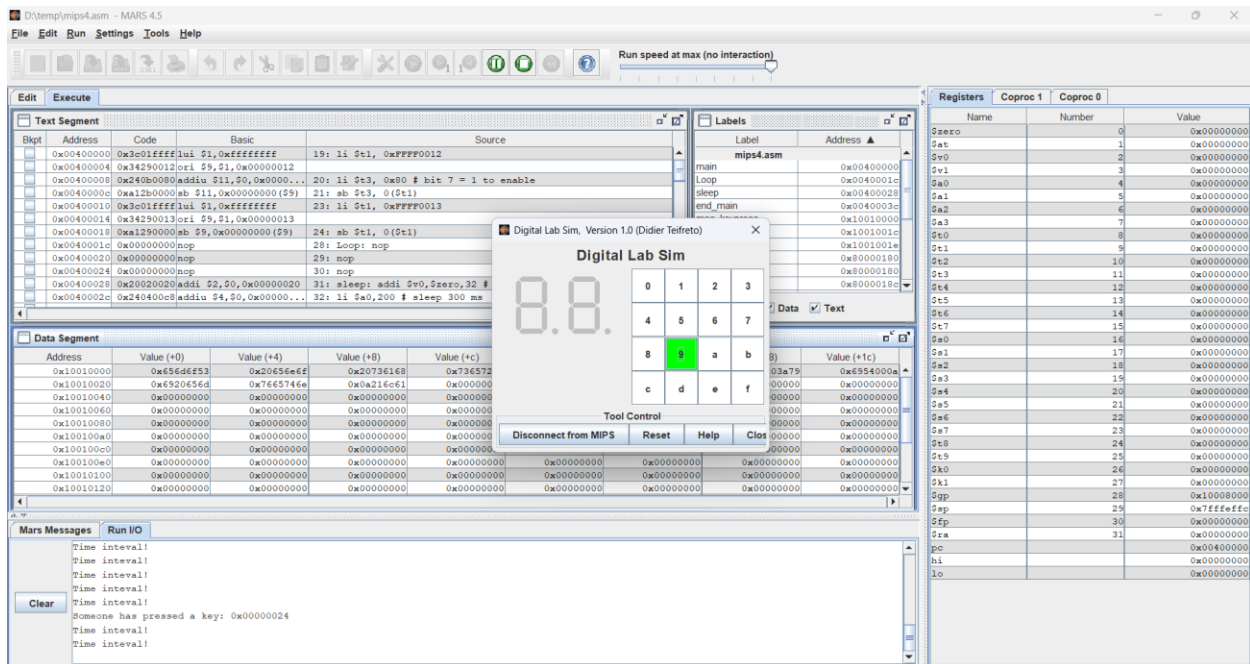
-Nếu là của Keyboard kiểm tra xem là bấm nút nào bằng cách lần lượt gán giá trị \$t1 lần lượt bằng 0x01, 0x02, 0x04, 0x08 để kiểm tra xem hàng nào được bấm. Load giá trị \$t2, nếu != 0 thì in ra. Sau khi in ra thì re-enable ngắt của Keyboard.

-Kết quả

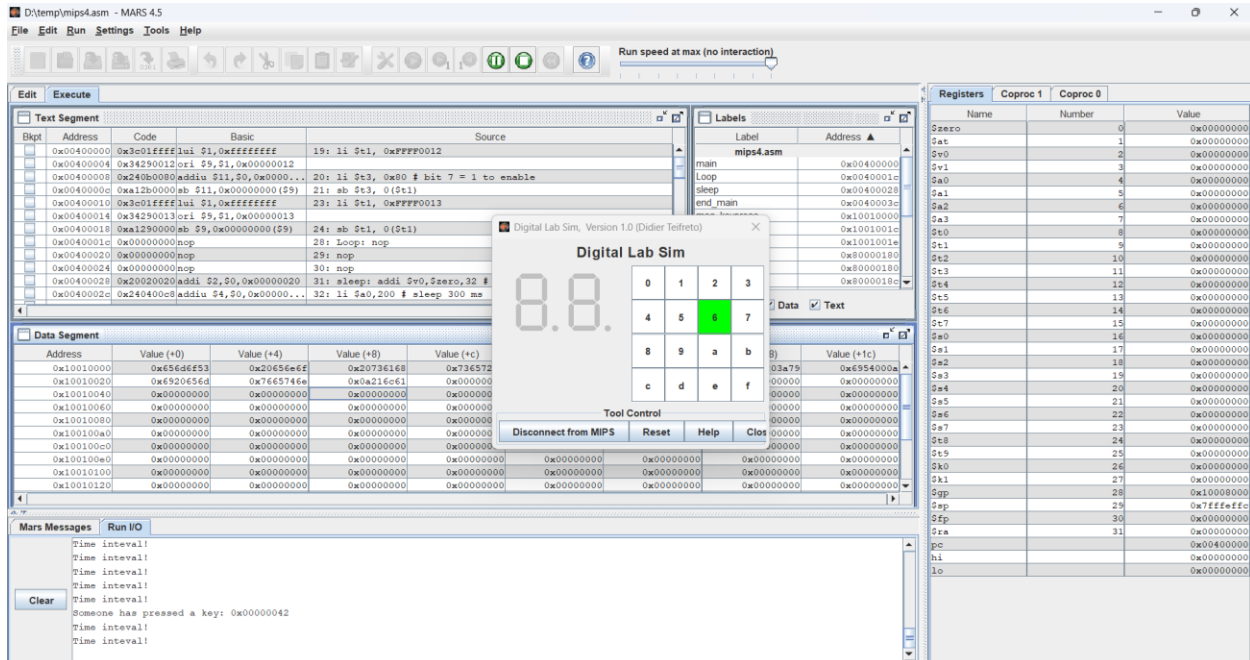
Bấm nút 4



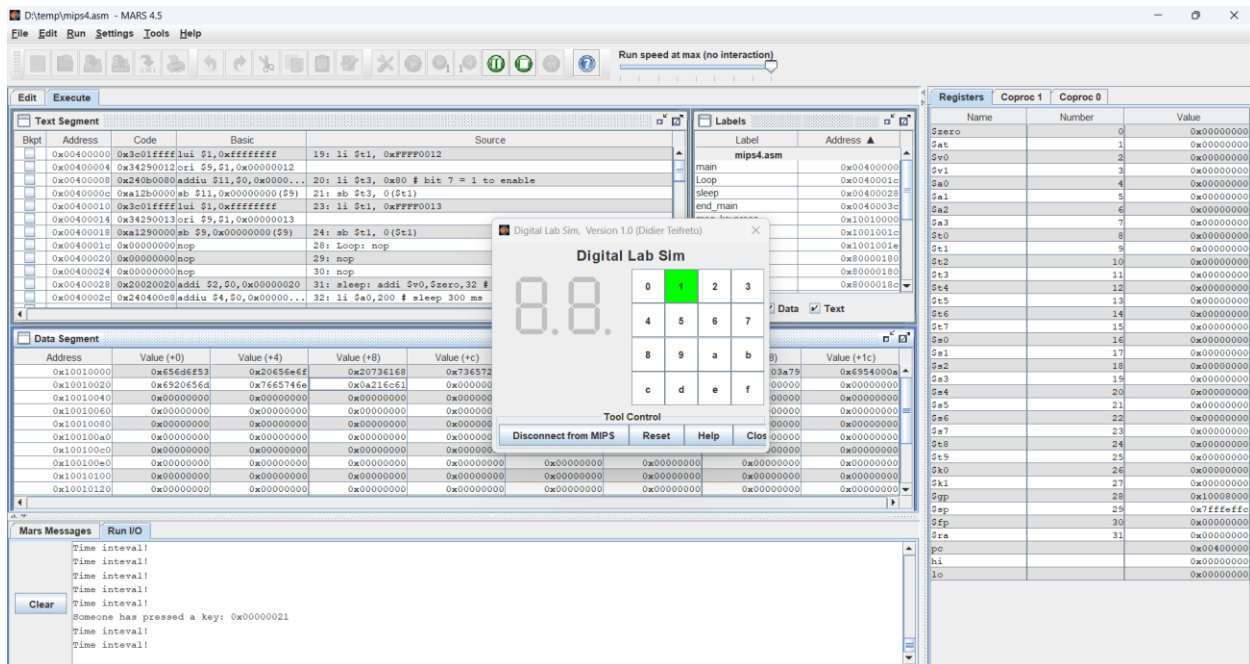
Bấm nút 9



Bấm nút 6



Bấm nút 1



Assignment 5

- Code

```
.eqv KEY_CODE 0xFFFF0004 # ASCII code from keyboard, 1 byte
.eqv KEY_READY 0xFFFF0000 # =1 if has a new keycode ?
# Auto clear after lw

.eqv DISPLAY_CODE 0xFFFF000C # ASCII code to show, 1 byte
.eqv DISPLAY_READY 0xFFFF0008 # =1 if the display has already to do
# Auto clear after sw

.eqv MASK_CAUSE_KEYBOARD 0x00000034 # Keyboard Cause

.text

li $k0, KEY_CODE
li $k1, KEY_READY
li $s0, DISPLAY_CODE
li $s1, DISPLAY_READY

loop: nop

WaitForKey: lw $t1, 0($k1) # $t1 = [$k1] = KEY_READY
beq $t1, $zero, WaitForKey # if $t1 == 0 then Polling
MakeIntR: teqi $t1, 1 # if $t0 = 1 then raise an Interrupt
j loop

#-----
# Interrupt subroutine
#-----

.ktext 0x80000180

get_caus: mfc0 $t1, $13 # $t1 = Coproc0.cause
IsCount: li $t2, MASK_CAUSE_KEYBOARD # if Cause value confirm Keyboard..
and $at, $t1, $t2
beq $at, $t2, Counter_Keyboard
j end_process
```

Counter_Keyboard:

ReadKey: lw \$t0, 0(\$k0) # \$t0 = [\$k0] = KEY_CODE

WaitForDis: lw \$t2, 0(\$s1) # \$t2 = [\$s1] = DISPLAY_READY

beq \$t2, \$zero, WaitForDis # if \$t2 == 0 then Polling

Encrypt: addi \$t0, \$t0, 0 # change input key

ShowKey: sw \$t0, 0(\$s0) # show key

nop

end_process:

next_pc: mfc0 \$at, \$14 # \$at <= Coproc0.\$14 = Coproc0.epc

addi \$at, \$at, 4 # \$at = \$at + 4 (next instruction)

mtc0 \$at, \$14 # Coproc0.\$14 = Coproc0.epc <= \$at

return: eret # Return from exception

- Thực thi

-Chương trình thực hiện kiểm tra nếu bàn phím được bấm thì nhảy vào đoạn mã cho xử lý ngắt bằng lệnh teqi.

-Ở đoạn mã xử lý ngắt, kiểm tra xem có phải ngắt của bàn phím hay không. Nếu phải thì show key đó ra màn hình.

-Kết quả

Keyboard and Display MMIO Simulator**DISPLAY: Store to Transmitter Data 0xffff000c, cursor 42, area 95 x 10**

Yjo!dibp!npj!ohvpj!ufo!upj!mb!Mf!Rvpd!Eboh

Font

☒ DAD

Fixed transmitter delay, select using slider

Delay length: 5 instruction executions

KEYBOARD: Characters typed here are stored to Receiver Data 0xffff0004

Xin chao moi nguoi ten toi la Le Quoc Dang

Tool Control

Disconnect from MIPS

Reset

Help

Close